

Archive It with Zenodo

DOIs for your data, materials, and code (with GitHub releases)

Erin M. Buchanan

October 6, 2026

Plan for Today

- Why archive, and what Zenodo is
- Key ideas: records, DOIs, versions, access
- Route 1: upload files directly (demo)
- Route 2: archive a GitHub release (demo)
- Tips for big teams
- Q&A, plus help with your own projects

Why Archive?

- Get credit for everything you make: data, materials, code, and protocols, not just papers
- Funders and journals increasingly ask for persistent identifiers (DOIs)
- OSF has changed, so many networks need a new home for files and DOIs
- Links break: websites and repos get renamed, moved, or deleted, but a DOI keeps working

What Is Zenodo?

- A general-purpose repository run by CERN
- Free to use, and the software is open source (InvenioRDM)
- Accepts any research output: data, code, slides, posters, protocols, preprints
- Up to 50 GB and 100 files per record
- Every record gets a DOI
- Log in at zenodo.org with ORCID, GitHub, or an email account

Key Idea: Records and Versions



- Each record is one thing you archive. Each version gets its own DOI.
- The concept DOI ties all the versions together.
- Need a change? Make a new version, not a new record.

See it in action: [jaspSyntheticData](#) has v0.1.0 and v0.2.0 under one concept DOI.

Key Idea: Locked Files, Editable Descriptions

Files: locked

After you publish, files can't be added, removed, or swapped. New files = new version.

Description: editable

Fix the title, authors, description, or links any time, with no new version needed.

Why that's good:

- Preregistrations and protocols get a permanent timestamp
- Every version stays exactly as it was, so changes are traceable
- The description grows with the project: add the paper, data, and code links later

Key Idea: Access Options

Choose who can download the files:

Open

Anyone can download the files

Restricted

People request access and you approve

Embargoed

Files open up automatically on a date you pick

The description is always public, even when the files aren't.

Share Links for Peer Review

Give reviewers access without making anything public:

Can view

Restricted or embargoed files
of a published record

Can preview draft

A record you haven't published
yet

Can edit

Collaborators who help fill in
the draft

1. Open your record or draft and click Share
2. Go to the Links tab
3. Optional: add a title and an expiration date
4. Pick the permission, then click Create a new link
5. Copy the link and send it (e.g., in your journal submission)

Delete the link from the same tab whenever you want to turn off access.

Practice First: The Sandbox

- sandbox.zenodo.org is a full copy of Zenodo just for testing
- Separate account; nothing there is permanent
- DOIs from the sandbox are not real

Route 1: Upload Directly

Upload: Basic Information

1. Go to My dashboard (top menu), then click New upload
2. Files: drag in your files
3. Basic information:
 - Digital Object Identifier: “Do you already have a DOI?” → No
 - Resource type: dataset, software, presentation...
 - Title and publication date
 - Creators: everyone in the citation, with ORCIDs
 - Description: what’s in the files and how to use them
 - Licenses: CC BY 4.0 by default. Help choosing: [Creative Commons chooser](#) or [choosealicense.com](#) for code

Upload: Recommended Info & Related Works

4. Recommended information:

- Contributors: helped, but not in the citation. Each needs a role (e.g., data collector)
- Keywords and subjects: your own, or from EuroSciVoc or MeSH
- Languages, version, and publisher

5. Related works: link the paper, preprint, preregistration, or repo

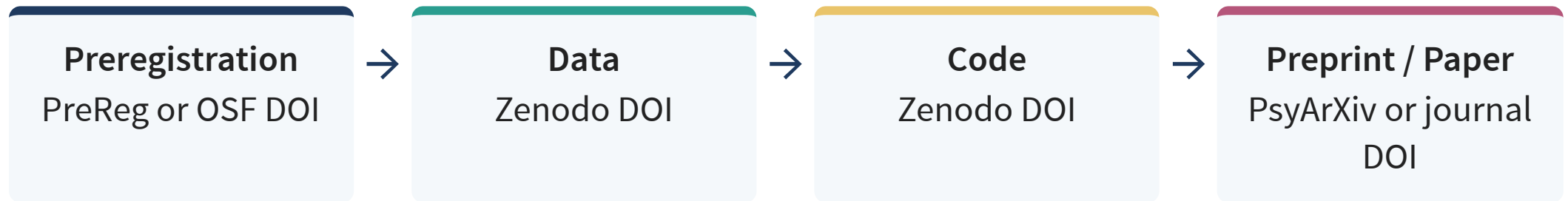
- Pick a relation (e.g., “Is supplement to”) and paste the DOI or URL

6. Choose access (open, restricted, embargoed), then Publish

Zenodo’s form has more sections too (funding, alternate identifiers, references, and more), but these are the most important ones to fill in.

Related Works: Connecting Your DOIs

One project, many DOIs. Related works tell readers (and search indexes) how they fit together:



This record...	Relation	...links to
Data	Is documented by	the preregistration
Data	Is supplement to	the preprint or paper
Code	Is supplement to	the data
Data v2	Is derived from	an earlier dataset

Add links as they appear: when the paper is published, edit the record and add its DOI. No new version needed.

Route 2: Archive a GitHub Release

Commits vs. Releases

Commit

A saved change to your files. You make lots of them, every time you work on the project.

Like hitting save on a document

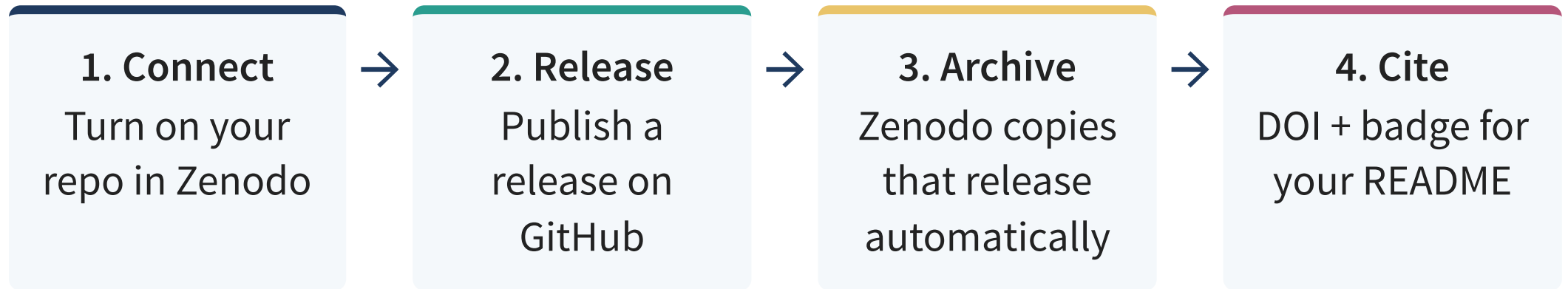
Release

A named snapshot of the whole repo at one moment (e.g., [v1.0.0](#)), with notes about what changed.

Like submitting a version of the paper

- Zenodo ignores commits. It only archives releases.
- Each release becomes a new version of your Zenodo record, with its own DOI.
- Make a release when something is done: data collection closes, the analysis is final, the paper is submitted.

How It Works



Every new GitHub release becomes a new version of the same Zenodo record.

Connecting GitHub, Step by Step

In Zenodo:

1. Log in, then open your profile menu → GitHub
2. Click Sync now
3. Find your repository and flip the switch to on

In GitHub:

4. Go to your repo → Releases → Draft a new release
5. Add a tag (e.g., **v1.0.0**), a title, and notes
6. Click Publish release

Back in Zenodo: wait a few minutes, then find your new DOI on the GitHub page.

Describe It: CITATION.cff

Add a `CITATION.cff` file to your repo so Zenodo (and GitHub) know how to cite it:

```
1 cff-version: 1.2.0
2 title: "Example BTS Project: Shared Analysis Code"
3 version: 1.0.0
4 license: "MIT"
5 type: software
6 message: "If you use this, please cite it as below."
7 authors:
8   - given-names: Erin M.
9     family-names: Buchanan
10    affiliation: Harrisburg University
11    orcid: "https://orcid.org/0000-0002-9689-4189"
12   - given-names: "[Next]"
13     family-names: "[Author]"
14 keywords:
15   - big team science
```

Bonus: GitHub shows a “Cite this repository” button.

Or: .zenodo.json

More control: contributors, funding, communities, and links to your paper. If both files exist, Zenodo uses only `.zenodo.json`.

```
1 {
2   "title": "Example BTS Project: Shared Analysis Code",
3   "upload_type": "software",
4   "license": "mit",
5   "creators": [
6     {"name": "Buchanan, Erin M.",
7       "affiliation": "Harrisburg University",
8       "orcid": "0000-0002-9689-4189"}
9   ],
10  "contributors": [
11    {"name": "Site Lead, Example", "type": "DataCollector"}
12  ],
13  "related_identifiers": [
14    {"identifier": "https://doi.org/[paper DOI]",
15     "relation": "isSupplementTo",
16     "resource_type": "publication-article"}
17  ],
18  "communities": [{"identifier": "[your-network]"}]
19 }
```

Live Demo: GitHub Release

github.com/doomlab/btscon-2026: the slides from this conference!

1. Switch the repo on in the Zenodo sandbox
2. Show `CITATION.cff` and `.zenodo.json`. This repo has both, so Zenodo uses `.zenodo.json`
3. Publish release `v1.0.0` on GitHub
4. Find the new record and DOI in Zenodo
5. Add the DOI badge to the README
6. Publish `v1.1.0` and watch a new version appear

When It Doesn't Work

- No record appeared? Check the repo was switched on *before* the release, and that the release was published (not a draft).
- Release failed? In Zenodo's GitHub page, click the release, then Errors. It's usually the metadata.
- Bad `CITATION.cff`? Validate it with `cffconvert` first.
- Wrong authors or title? Fix the metadata file, then publish a new release.

Tips for Big Teams

Zenodo for Big Team Science

- Make a Zenodo Community for your network to collect every project's outputs in one place (example: [CREP's community](#))
- Get everyone's ORCID before you publish, because author lists are long
- Link everything with related identifiers: preprint ↔ data ↔ code ↔ materials
- One record per output, with new versions when things change
- Use restricted access for sensitive data so it's citable but controlled
- Add your funder (e.g., SSHRC grant) in the metadata

Cheat Sheet

Zenodo	zenodo.org
Practice	sandbox.zenodo.org
Help docs	help.zenodo.org
GitHub integration	help.zenodo.org/docs/github
Make a CITATION.cff	cff-initializer
GitHub releases refresher	Pre-conference workshop recording

Questions?

Let's archive something!